

Hollow Core Fibres: From Visionary Concept to Multi-Problem Solution

Francesco Poletti^(1,2)

⁽¹⁾ Optoelectronics Research Centre, University of Southampton, UK, fp@soton.ac.uk

⁽²⁾ Microsoft Azure Fiber, Romsey, UK

Abstract *Hollow-core fibres have evolved from early photonic bandgap concepts to modern antiresonant nested designs, overcoming loss, bandwidth, and scalability challenges. Here we briefly trace this progress, highlighting recent DNANF advances achieving sub-silica loss, long spans, and multi-band operation: ©2026 The Author(s)*

Introduction

The concept of guiding light in a hollow core, where the optical field propagates predominantly in air rather than glass, has long represented an attractive but elusive goal in optical communications [1]. By removing the overlap of light with the material, hollow core fibres (HCFs) promise a unique combination of ultra-low nonlinearity, reduced latency, low dispersion and, ultimately, the prospect of transmission loss below the fundamental limits imposed by Rayleigh scattering in silica. Over more than two decades, this vision has evolved through multiple technological revolutions, from the early photonic bandgap fibres (PBGFs) to today's highly optimised antiresonant nested structures, culminating in optical fibres that not only rival but surpass the performance of solid-core single-mode fibres (SMFs) across broad spectral regions.

This invited talk will provide a historical overview of the key scientific and technological breakthroughs that have enabled HCFs to transition from a laboratory curiosity into a compelling alternative platform for optical transmission. Particular emphasis will be placed on how successive design paradigms addressed the dominant physical loss mechanisms and engineering constraints, ultimately transforming hollow core fibre technology into a versatile “multi-problem solution” for modern optical networks.

Early Developments: Photonic Bandgap Guidance

The first practical demonstration of hollow core fibre guidance was achieved at the turn of the century using photonic bandgap structures, where light is confined in an air core by a periodic microstructured cladding. Early advances rapidly pushed attenuation from tens of dB/km down to the 1–2 dB/km regime, with a landmark result reporting 1.7 dB/km loss near 1550 nm [2]. This proof-of-principle demonstration was pivotal: it showed that long-distance guidance in air was possible and that the dominant material absorption and nonlinearity mechanisms of glass could in principle be mitigated. However, photonic

bandgap fibres suffered from intrinsic limitations that ultimately constrained their scalability toward telecommunications applications. In particular, their relatively narrow operational bandwidths, sensitivity to structural perturbations, loss mechanisms dominated by surface scattering and insufficient modal purity limited further progress. While these fibres established the feasibility of hollow core guidance, they also highlighted that a fundamentally different guiding mechanism would be required to achieve the performance levels necessary for practical deployment.

The Antiresonant Paradigm and the Emergence of NANF

A decisive shift occurred with the introduction of antiresonant hollow core fibres (AR-HCFs), in which light confinement is achieved through inhibited coupling and antiresonant reflection from thin glass membranes surrounding the core [3,5]. This guiding mechanism dramatically reduces the fraction of optical power interacting with glass, thereby suppressing surface scattering loss as compared to PBGF.

The development of Nested Antiresonant Nodeless Fibres (NANFs) represented a major milestone in this new paradigm [6]. By introducing nested glass tubes in the cladding, NANFs enhanced confinement while crucially also simultaneously enabling strong suppression of higher-order modes. The first low-loss NANFs demonstrated attenuation as low as **1.3 dB/km** together with the first successful data transmission experiments [7], followed shortly by further reductions to **0.65 dB/km across the C and L bands** [8].

These results marked a critical transition: hollow core fibres were no longer merely scientific curiosities but had become viable transmission media. Importantly, the dominant loss mechanisms had shifted from surface scattering to a combination of confinement (or leakage) and microbend effects, indicating that further improvements could be pursued through structural optimisation and fabrication control.

Rapid Loss Reduction and Understanding of Limiting Mechanisms

From 2020 onwards, a series of advances in design, fabrication uniformity and measurement methodology led to an unprecedented acceleration in performance improvements. NANFs achieved losses of ~ 0.28 dB/km, approaching those of conventional SMFs [9]. Crucially, this phase also brought a deeper understanding of measurement artefacts, such as the influence of higher-order modes on short cut-back measurements, enabling more accurate determination of the true fibre attenuation.

Further optimisation of the fibre geometry, including modifications to tube number and arrangement (e.g. from 6-tube to 5-tube NANFs), enabled record losses of ~ 0.22 dB/km while simultaneously improving higher-order mode suppression and extending operation across a wide wavelength range, from the near-infrared telecom bands down to the visible [10]. Notably, these results demonstrated for the first time that hollow core fibres could outperform silica fibres at wavelengths where Rayleigh scattering fundamentally limits solid-core performance.

This period established a key conceptual shift: rather than being constrained by material properties, the performance of optical fibres (with a hollow core) became primarily governed by geometry and fabrication precision.

Double-Nested Structures and the Achievement of Silica-Equivalent Loss

The introduction of Double Nested Antiresonant Nodeless Fibres (DNANFs) represented the next major leap forward. By adding an additional nested layer to enhance confinement, DNANFs significantly reduced leakage loss while allowing smaller core sizes, thereby also mitigating microbend sensitivity.

This design enabled the first hollow core fibres to match the loss of conventional silica SMF, achieving approximately 0.174 dB/km in the C-band. This result marked a historic milestone: after decades of research, hollow core fibres had reached parity with the incumbent technology, whilst providing lower latency and nonlinearity [11].

Breaking the Silica Limit and Addressing Scalability

Subsequent work pushed performance beyond this threshold. Through systematic optimisation of geometry and fabrication processes, DNANFs achieved losses below 0.1 dB/km, with records of ~ 0.08 dB/km at 1550 nm, the lowest ever reported in any optical fibre up to that point. At this stage, leakage contributions had been reduced to

a minor component, and loss was dominated by surface scattering and microbending, indicating that the fundamental mechanisms limiting performance had changed once again [12].

In parallel, new structural concepts such as interstitial-tube-assisted DNANFs addressed the critical challenge of manufacturability. By relaxing structural tolerances, these fibres achieved both record low loss (≈ 0.05 dB/km) and unprecedented single-draw lengths exceeding 80 km, demonstrating for the first time a credible path toward industrial scalability [13].

Beyond Loss: Towards Spectral and Functional Engineering

More recent developments highlight a further evolution in the role of hollow core fibres. Rather than focusing solely on minimising loss over a single wide spectral band covering S+C+L wavebands, new designs aim to engineer multiple transmission windows and broader functionality. An example is the introduction of hybrid multi-window DNANFs, capable of delivering simultaneously ultra-low loss at both ~ 1550 nm and ~ 1 μ m, with losses of approximately 0.11 dB/km and 0.13 dB/km, respectively [14].

This capability enables entirely new classes of applications, including simultaneous data transmission and high-power laser delivery, exploitation of alternative amplifier bands, and hybrid optical systems combining communication and sensing functionalities. Such developments illustrate how HCFs have evolved from a single-problem solution (reducing nonlinearity or latency) into a platform technology capable of addressing multiple system-level challenges.

Conclusions and Outlook

The evolution of hollow core fibre technology over the past two decades can be understood as a sequence of breakthroughs addressing a succession of fundamental limitations. Early work established the feasibility of guiding light in air, while subsequent advances in antiresonant designs delivered the bandwidth, robustness, and modal control required for practical transmission. The introduction of increasingly sophisticated nested structures enabled dramatic reductions in loss, culminating first in parity with conventional silica fibres and, more recently, in performance that surpasses the fundamental attenuation limits of glass.

As a result of these developments, many of the longstanding challenges that once hindered the practical adoption of hollow core fibres have now been largely overcome. Loss has been reduced to levels below those achievable in solid-core fibres, modal behaviour is well controlled,

and advances in design and fabrication have improved both structural uniformity and scalability. At the same time, new fibre architectures demonstrate compatibility with existing infrastructure and open the possibility of engineering transmission properties across multiple spectral bands.

Taken together, these achievements highlight a profound shift in the role of hollow core fibres. Rather than representing an alternative technology addressing a single limitation, such as non-linearity or latency, they now emerge as a comprehensive platform capable of tackling several key constraints of modern optical systems simultaneously. While further work is needed to fully realise large-scale manufacturing and deployment, the trajectory of progress clearly indicates that hollow core fibres are poised to play a significant role in the future of optical communications, offering a compelling and versatile complement to, and in some cases a replacement for, conventional single-mode fibres.

References

- [1] Philip St. J. Russell, "Hollow-core photonic crystal fibres," *Nature*, Vol. 424, pp. 674–677, 2003.
- [2] B. J. Mangan, L. Farr, A. Langford, P. J. Roberts, D. P. Williams, F. Couny, M. Lawman, M. Mason, S. Coupland, R. Flea, H. Sabert, T. A. Birks, J. C. Knight, and P. St. J. Russell, "Low loss (1.7 dB/km) hollow core photonic bandgap fibre," *Optical Fiber Communication Conference (OFC 2004)*, Postdeadline Paper PDP24.
- [3] F. Benabid, J. C. Knight, G. Antonopoulos, and P. St. J. Russell, "Stimulated Raman scattering in hydrogen-filled hollow-core photonic crystal fibre," *Science*, Vol. 298, pp. 399–402, 2002.
<https://doi.org/10.1126/science.1076408>
- [4] Y. Y. Wang, F. Couny, F. Benabid, and P. St. J. Russell, "Low loss broadband transmission in hollow-core photonic crystal fiber with hypocycloid core contour," *Optics Letters*, Vol. 36, No. 5, pp. 669–671, 2011.
<https://doi.org/10.1364/OL.36.000669>
- [5] A. N. Kolyadin et al., "Light transmission in negative curvature hollow core fibres," *Optics Express*, Vol. 21, No. 8, pp. 9514–9519, 2013.
<https://doi.org/10.1364/OE.21.009514>
- [6] Francesco Poletti, "Nested antiresonant nodeless hollow core fiber," *Optics Express*, Vol. 22, No. 20, pp. 23807–23828, 2014.
<https://doi.org/10.1364/OE.22.023807>
- [7] T. D. Bradley, J. R. Hayes, Y. Chen, G. T. Jasion, S. R. Sandoghchi, R. Slavík, E. N. Fokoua, S. Bawn, H. Sakr, I. A. Davidson, A. Taranta, J. P. Thomas, M. N. Petrovich, D. J. Richardson, and F. Poletti, "Record low-loss 1.3 dB/km data transmitting antiresonant hollow core fibre," *European Conference on Optical Communication (ECOC 2018)*, Postdeadline Paper Th3F.2.
- [8] T. D. Bradley, G. T. Jasion, J. R. Hayes, Y. Chen, L. Hooper, H. Sakr, M. Alonso, A. Taranta, A. Saljoghei, H. C. Mulvad, M. Fake, I. A. Davidson, N. V. Wheeler, E. N. Fokoua, W. Wang, S. R. Sandoghchi, D. J. Richardson, and F. Poletti, "Antiresonant hollow core fibre with 0.65 dB/km attenuation across the C and L telecommunication bands," *European Conference on Optical Communication (ECOC 2019)*, Postdeadline Paper Th3F.1.
- [9] G. T. Jasion, T. D. Bradley, K. Harrington, H. Sakr, Y. Chen, E. N. Fokoua, I. Davidson, A. Taranta, J. R. Hayes, D. J. Richardson, and F. Poletti, "Hollow core NANF with 0.28 dB/km attenuation in the C and L bands," *Optical Fiber Communication Conference (OFC 2020)*, Postdeadline Paper Th4B.4.
- [10] Hesham Sakr et al., "Hollow core NANFs with five nested tubes and record low loss at 850, 1060, 1300 and 1625 nm," *Optical Fiber Communication Conference (OFC 2021)*, Postdeadline Paper F3A.4.
- [11] Gregory T. Jasion, Hesham Sakr, John R. Hayes, Seyed Reza Sandoghchi, Lucy Hooper, Eric N. Fokoua, Arsalan Saljoghei, Hans Christian Mulvad, Marcelo Alonso, Austin Taranta, Thomas D. Bradley, Ian A. Davidson, Yong Chen, David J. Richardson, and Francesco Poletti, "0.174 dB/km hollow core double nested antiresonant nodeless fiber (DNANF)," *Optical Fiber Communication Conference (OFC 2022)*, Postdeadline Paper Th4C.7.
- [12] Y. Chen, M. N. Petrovich, E. N. Fokoua, A. I. Adamu, M. R. A. Hassan, H. Sakr, R. Slavík, S. Bakhtiari Gorajooobi, M. Alonso, R. Fatobene Ando, A. Papadimopoulos, T. Varghese, D. Wu, M. Fatobene Ando, K. Wisniowski, S. R. Sandoghchi, G. T. Jasion, D. J. Richardson, and F. Poletti, "Hollow core DNANF optical fiber with <0.11 dB/km loss," *Optical Fiber Communication Conference (OFC 2024)*, Postdeadline Paper Th4A.8.
- [13] Shoufei Gao, Hao Chen, Yifan Xiong, Yizhi Sun, Wei Ding, and Yingying Wang, "Interstitial-tube-assisted hollow-core fibre with 0.052 dB/km loss and 83 km span," *European Conference on Optical Communication (ECOC 2025)*, Postdeadline Paper Th03.01.1.
- [14] Ghafour Amouzad Mahdiraji, Seyed Mohammad Abokhamis Mousavi, Dong Wu, Thejus Varghese, Naveen Krishna Baddela, Muhammad Rosdi B. Abu Hassan, Zahra Kakaei, Abubakar Isa Adamu, Ziwei Zhai, Krystian Wisniowski, Shahab Bakhtiari Gorajooobi, Ali Shakiba, Mahmudur Rahman, Jaroslaw Rzegocki, Gianluca Guerra, Eric N. Fokoua, Yong Chen, Gregory Jasion, and Francesco Poletti, "First dual-band hybrid window antiresonant hollow-core fiber with 0.13 dB/km loss at 1 μm and 0.11 dB/km at 1.55 μm ," *Optical Fiber Communication Conference (OFC 2026)*, Postdeadline Paper Th4B.8.